

SCOPE OF WORK & PROJECT DELIVERABLES

Official Proposal, Technical Scope Matrix & Platform Architecture Document

ORGANIZATION	HEADQUARTERS	REGULATORY STATUS	DOCUMENT REF
RevDrones Technologies Pvt. Ltd.	Vadodara, Gujarat, India	DGCA Compliant / Digital Sky	RD-SOW-2026-V3

01 Commercial Drone Operations Scope of Work

RevDrones provides end-to-end aerial data acquisition, crop spraying, and industrial non-destructive testing (NDT) across four specialized divisions:



1. Precision Agriculture & Crop Intelligence

HIGH-PAYLOAD SPRAYERS

- Airspace & Terrain Mapping:** Pre-flight topographic scanning, boundary demarcation, and obstacle elevation profiling.
- Autonomous High-Capacity Spraying:** Uniform droplet dispersal (micron-level atomization) for liquid fertilizers, bio-nutrients, and pesticides, covering up to 40–50 acres/day with zero crop trampling and 90% water conservation.
- Multispectral Crop Health Analysis:** NDVI & NDRE aerial imaging to identify localized pest infestations, nitrogen deficiency, and soil moisture variations.
- Variable-Rate Prescription (VRA) Maps:** Geo-referenced application zone mapping for targeted input distribution.



2. Aerial Surveying, GIS & Topographic Mapping

SURVEY-GRADE RTK/PPK

- Ground Control Points (GCPs):** DGPS/RTK ground survey marker deployment ensuring absolute spatial accuracy ($\leq 2\text{--}5\text{ cm}$).
- Photogrammetric & LiDAR Flights:** High-overlap nadir/oblique imagery acquisition using calibrated full-frame sensors and LiDAR.
- Topographic & Elevation Modeling:** Production of High-Res 2D Orthomosaics, Digital Surface Models (DSM), Digital Terrain Models (DTM), and 0.5m/1.0m contour intervals.
- Volumetric & Cadastral Analysis:** Cut-and-fill volumetrics for mining stockpiles, quarry pits, road/rail corridor alignments, and land title records.



3. Industrial Visual & Radiometric Thermal NDT Inspection

CRITICAL ASSET SAFETY

- High-Risk Structural Inspection:** 4K optical zoom inspection of flare stacks, cooling towers, chimney flues, storage tanks, and bridges without hazardous scaffolding or human rope access.
- Solar PV Farm Diagnostics:** Panel-level thermal mapping to detect hot spot cells, string anomalies, bypass diode faults, and soiling accumulation.
- Power Transmission Line Corridors:** Component-level defect detection (damaged insulators, conductor strand breakage, tower corrosion) and right-of-way (ROW) vegetation clearance audits.
- Pipeline Corridors:** Thermal infrared signature analysis for subsurface fluid/gas leaks, coating degradation, and unauthorized right-of-way encroachment.

⚡ 4. Revolutionized SaaS AI Analytics Platform AUTOMATED ANALYTICS

- **AI Defect Recognition:** Automated detection and classification of wind blade cracks, erosion, lightning strikes, solar thermal anomalies, and powerline faults.
- **Severity Prioritization:** Automated defect ranking (Critical, Moderate, Low) tagged with millimeter-accurate GPS coordinates.
- **Interactive Cloud Access:** Multi-user cloud portal for CAD layout overlay, historical comparison scans, and 1-click PDF repair queues.

02 What We Provide (Tangible Project Deliverables)

Every engagement produces engineering-grade data delivered in standard industrial and GIS formats:

Service Vertical	Deliverables Provided to Client	Standard Formats
Precision Agriculture	• High-Resolution Orthomosaic Farm Map • NDVI/NDRE Crop Health Index Maps • Variable-Rate Prescription (VRA) Maps • Mission Spray Completion & Volume Logs	GeoTIFF, Shapefile (SHP), PDF, CSV
Surveying & GIS	• Survey-Grade 2D Orthomosaic ($\leq 2\text{cm/pixel}$ GSD) • 3D Dense Point Clouds & 3D Mesh • Digital Surface (DSM) & Terrain (DTM) Models • Topographic Contour Drawings & Boundary Maps • Stockpile Volume Calculation Reports	GeoTIFF, LAS / LAZ, DXF / DWG, SHP, PDF, KMZ
Industrial Inspection	• High-Res Visual RGB Image Catalog • Radiometric Thermal (FLIR) Image Datasets • Engineering Defect Audit Report with Severity Ranking • GIS Defect Overlay on Site CAD Layout	Annotated PDF, Raw Radiometric JPEG, CAD DXF, KMZ
SaaS Intelligence	• Web Portal Access for interactive asset inspection • Automated Defect Summary & Historical Health Index • Exportable Actionable Work Orders & Repair Queues	Web Cloud Portal, PDF, Excel/CSV

03 End-to-End Operational Execution Process

Our standard operating procedure (SOP) ensures regulatory safety, mission precision, and rapid data turnaround:



04 RevDrones Web Platform & Application Scope

The RevDrones web platform serves as the digital flagship, client booking engine, and back-office management console:

1. Client-Facing Frontend & Interactive Experience NEXT.JS 14 & THREE.JS

- **Interactive 3D Aerospace Diorama:** WebGL/Three.js interactive 3D hero scene with realistic drone telemetry and flight vectors.
- **Service & Industry Hubs:** 9 specialized landing pages detailing Agriculture, Survey, Inspection, and 6 industry sectors.
- **SaaS Platform Portal:** 4 dedicated 3D interactive sector scenes (Wind, Solar, Powerline, Pipeline) with real-time HUD telemetry overlays.
- **Dynamic Blog Hero Carousel:** 5-second auto-swipe carousel matching client reviews, dynamically updated from Supabase.
- **Client Review Carousel:** Smooth auto-sliding testimonial showcase with avatar fallbacks and pause-on-hover.

2. 24/7 Instant AI Flight Assistant (Chatbot) SUB-SECOND RESPONSES

- **High-Tech Animated Drone Launcher:** Custom vector drone with spinning propeller blades, pulsing AI core LED, radar ping waves, and floating prompt tooltip (⚡ *Ask Flight AI*).
- **Comprehensive Knowledge Base:** Sub-second answers for drone services, hardware specs, flight ranges (15km+), weather constraints, pricing consultations, booking steps, and office coordinates.
- **Interactive Interface:** Quick suggestion chips, clickable markdown links to website forms, and chat history management.

3. Staff Operations Suite & Real-Time Dashboard SUPABASE BACKEND

- **Authentication & Session Guard:** Secure staff login (/login) with automatic session verification and redirection.
- **Overview KPIs & Signals:** 4 live stat cards (Total Bookings with 7-day sparklines, Pending Requests, Weekly Leads, Published Blogs), 7-day booking signal bar graph, and activity feeds.
- **Live Bookings Management:** Searchable client booking table with live status dropdowns (*pending, confirmed, completed, cancelled*) updating Supabase in real-time.
- **Blog & Reviews CMS:** Full content management suite to create, edit, auto-generate slugs, publish, reorder, and delete articles and testimonials.
- **Dual-Action Dispatch:** Submissions save directly to Supabase and simultaneously trigger instant email alerts via Google Apps Script Webhooks.

05 Safety, Compliance & Certifications

- **DGCA Certified Remote Pilots:** Every mission is commanded by a certified Remote Pilot-in-Command (RPIC).

- **Digital Sky Compliance:** Strict adherence to Indian civil aviation airspace regulations (Green, Yellow, and Red zones).
- **Aviation Liability Coverage:** Comprehensive third-party and equipment insurance policies.
- **Fail-Safe Protocols:** Triple-redundant Return-to-Home (RTH), geo-fencing limits, and terrain-following sensors.

Ready to discuss your project?

Contact our operations team for custom flight planning, spatial resolution specifications, or quotation requests.

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